UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2010 question paper

for the guidance of teachers

5054 PHYSICS

5054/41 Paper 4 (Alternative to Practical), maximum raw mark 30

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

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UNIVERSITY of CAMBRIDGE
International Examinations

1	**(a)**	152 202 252 303	B1	[1]
	(b)	axes, correct way round, labelled quantity and unit	B1	
		y: 2 cm ≡ 50 g *x*: 2 cm ≡ 50 s or 80 s		
		scales: more than ½ page, sensible	B1	
		points plotted accurately	B1	
		best fit straight line neatly drawn	B1	[4]
	(c)	directly proportional / doubling *x* doubles *y*	B1	[1]
	(d)	0.84 ± 0.02 ignore unit	B1	[1]
	(e)	2400 ± 50 e.c.f. 2020/(d)	B1	[1]
	(f)	1460 – 860 or 600 seen	C1	
		715 ± 15 s e.c.f. 600/(d)	A1	[2]
	(g)	high temperature / kettle hot / may burn you	B1	
		changes reading on balance / measurement	B1	[2]
	(h)	*L* increases plus		
		as more water to be boiled away / less mass is boiled off / *m* decreases		
		gradient of graph is reduced	B1	[1]
				[Total: 13]
2	**(a)**	5 to 30 s unit required	B1	[1]
	(b)	**(i)** 1 trundle wheel / long tape measure / metre rules (at least two)		
		2 stopwatch / watch with second hand **both required**	B1	[1]
		(ii) timing several / at least 3 intervals	B1	
		intervals round the track marked/used	B1	
		each measures time from start	B1	[3]
	(c)	**(i)** initial curve in correct direction	B1	
		straight line	B1	[2]
		(ii) second graph line: gradient increasing then decreasing/		
		S-shaped curve	B1	
		lines meet at 100 m	B1	[2]
				[Total: 9]

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3	**(a)**	**(i)**	A	M0	
		(ii)	22Ω cao	A1	[1]
	(b)	**(i)**	circuit with resistor and d.c. power supply	M0	
			ammeter connected in series with resistor	A1	
			voltmeter connected in parallel with resistor	A1	[2]
		(ii)	tolerance / differences in manufacture / not all identical / student errors	B1	[1]
					[Total: 4]
4	**(a)**	**(i)**	rod drawn alongside right hand side of A	B1	[1]
		(ii)	end of rod A moves away / rotates can be shown on diagram	B1	[1]
	(b)		to be sure we get same sign charge / effect on each rod / ensure rods repel	B1	[1]
	(c)		charge is dissipated / leaks away into air / bench / lost / decreases	B1	[1]
					[Total: 4]

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